STRATEGIC PRODUCT PLACEMENT ANALYSIS

Comprehensive Final Project Report

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Team ID	SWTID-2026-4978
Project Title	Strategic Product Placement Analysis
Documentation Scope	Full Document Integration (Sections 1 through 11)

1. INTRODUCTION

1.1 Project Overview

In modern physical retail environments, maximize profitability relies heavily on quantitative inventory floor allocation rather than basic visual merchandising assumptions. The Strategic Product Placement Analysis project focuses on building an operational intelligence framework to decode hidden layout performance metrics. By ingesting storefront structural transactional records, the analytical engine correlates micro-level spatial placement with overall customer purchasing frequencies.

1.2 Purpose

The primary purpose of this deployment is to isolate key commercial revenue drivers across brick-and-mortar setups. The engine translates raw datasets into clean spatial trends, helping category managers evaluate actual floor margin elasticities. This removes historical blind spots from floor scheduling and establishes a data-driven model for high-margin shelf allocation.

2. IDEATION PHASE

2.1 Problem Statement

Standard physical storefront operations often experience sub-optimal profit margins because they rely on generalized visual patterns rather than hard transaction analysis. Without linking product categories to localized path layouts, stores lack visibility into real traffic behavior and product position metrics. This results in misallocated shelf space and uncaptured revenue across high-traffic zones.

2.2 Empathy Map Canvas

Operational design focused heavily on the daily challenges faced by store floor managers and inventory coordinators. Ground-level assessments revealed that team members are often overwhelmed by cluttered, unstructured logs while under pressure to hit target revenue coefficients. The visualization architecture is explicitly tailored to streamline these workflows, turning complex tables into intuitive grid maps.

2.3 Brainstorming

Initial design iterations evaluated multiple analytical frameworks to map checkout velocity against layout coordinates. The team prioritized building a modular structure capable of handling multi-category transactional profiles. This allows for scalable data processing without system lag during high-volume query periods.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Consumer path distributions were analyzed to model step-by-step browsing sequences across core departments. By tracking interaction points from entrance pathways to standard checkout lanes, the system maps localized demographic preferences against overall purchase sizes.

3.2 Solution Requirement

The technical core requires an analytics layout that can handle large data files, clean data types automatically, manage global multi-select filters, and host cross-sheet logic scripts without data dropouts or script errors.

3.3 Data Flow Diagram

Data processing follows a clean three-tier architectural pipeline: raw data ingestion through flat source files, metadata refinement via custom logical calculation schemas, and dashboard presentation utilizing real-time interactive coordinate matrices.

3.4 Technology Stack

The system is built on a highly reliable corporate business intelligence infrastructure, utilizing Tableau desktop data engines for advanced calculations, combined with Python preprocessing libraries to handle flat structural csv data transfers cleanly.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The system bridges operational gaps by providing direct visual correlation models. Category leaders can instantly cross-reference spatial positions against localized volume performance coefficients, turning raw numbers into clear, actionable shelf layout updates.

4.2 Proposed Solution

The platform introduces an interactive workspace dashboard that merges individual treemaps, pricing bar indices, demographic donuts, and campaign matrix cards into a single responsive network for holistic storefront space planning.

4.3 Solution Architecture

Built upon a multi-layered metadata pipeline, the solution ensures decoupled workbook execution where active data variables feed independent calculation panels natively, protecting system stability during large dataset operations.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Development was executed across strict, milestone-tracked phases: initial schema drafting, transactional pipeline data validation, visualization workspace grid composition, and final multi-page presentation design validation.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Project teams compiled the following validation matrix inside the model performance testing template structure:

S.NO.	PARAMETER	SCREENSHOT / VALUES
1.	**Data Rendered**	Successfully processed and rendered the raw transactional log file `Product Positioning.csv` into the engine environment. The imported data architecture was fully verified at exactly 10 distinct attribute fields and 1,000 structural transaction rows with zero value dropouts, row omissions, or loading execution faults.
2.	**Data Preprocessing**	Configured and forced explicit strict metadata type definitions for all relational variables. Explicitly locked currency figures as decimal floats (e.g., Price and Competitor's Price), parsed localized product position data fields as qualitative string indices, and mapped structural categorical fields natively to isolate null string variances.
3.	**Utilization of Filters**	Deployed three interconnected multi-select global action filters (Product Category, Product Position, and Consumer Demographics) across the active reporting sheets. Slicing tests confirm instant, real-time recalculations and dynamic visualization updates across all open views without visualization rendering lag.
4.	**Calculation Fields Used**	Programmed and verified dedicated computed metrics and logic calculations directly inside the workspace layout. These calculated dimensions evaluate variance distributions and lift thresholds, isolating physical store placement performance trends from background marketplace noise variables.
5.	**Dashboard Design**	No. of Visualizations / Graphs: 4 core integrated layout sheets compiled neatly onto a single dashboard container canvas (incorporating a hierarchical Spatial Floor Treemap, a multi-variable Demographics Donut chart, a Competitor Price Bar, and a Promotional Matrix Grid). Layout scaling tests verified clear visualization alignment and card rendering symmetry.
6.	**Story Design**	No. of Visualizations / Graphs: 8 multi-page narrative storyboards sequenced logically across interactive presentation cards. Story testing verified that narrative navigation links accurately transition between

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		historical benchmarks, demographic splits, and final placement recommendations.

7. RESULTS

7.1 Output Screenshots Summary

Operational execution generated clear empirical proof confirming the massive business leverage driven by floor positioning parameters. Over the 1,000 transaction cycles, high-exposure placements (e.g., End-cap arrays) generated predictable volume lift coefficients across buyer groups. Demographic charts cleanly break down volume clusters across target personas (Seniors, Students, and Families), providing reliable baselines for targeted physical floor layouts.

8. ADVANTAGES & DISADVANTAGES

Advantages:

- **Granular Margin Visibility:** Connecting position data with revenue metrics uncovers high-value display insights, helping stores eliminate historical spatial blind spots.
- **Lag-Free Slicing:** Optimized database indexing allows category managers to slice information instantly across departments without computing bottlenecks.

Disadvantages:

- **Batch Update Constraints:** System currently reads historical files and lacks an active data link to update layouts based on real-time daily traffic anomalies.
- **High-Level Categories:** Broad testing definitions (Food, Clothing, Electronics) mask micro-level variances among individual product SKUs.

9. CONCLUSION

This space analysis project successfully delivered a robust decision-support system tailored for retail optimization. By cleaning data tables, mapping explicit data attributes, and applying global interaction logic, the solution proves that strategic display management shapes consumer transaction trends. This establishes a scalable, data-driven approach for physical space planning.

10. FUTURE SCOPE

Future development will shift the system from historical batch tracking into automated live tracking. Roadmap priorities include building active API pipelines to integrate store sensor data, implementing automated forecasting scripts to predict seasonal sales shifts, and diving deeper into customer loyalty metrics for hyper-localized layout planning.

11. APPENDIX

This section provides direct download access and folder coordinates compiled across all active infrastructure testing phases.

Dataset Storage Repository Link:
`https://drive.google.com/drive/folders/1_e5gMhelFLk3wqlR8aQRclX8L4bO6-JQ`

Project Source Code & Repository Link:
`https://github.com/SWTID-2026-4978/strategic-product-placement-analysis`